Paper 2, Theorem 1.2: Statement Amendment

Internal research note for the `ShenWork` formalization

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Abstract

The parameter range printed in Theorem 1.2 of Chen–Ruau–Shen permits the mixed undamped branch $a > 0$, $b = 0$. In that branch a positive spatially constant solution grows exponentially, contradicting the asserted uniform boundedness. This note records the counterexample, locates the missing hypothesis in the published proof, gives two corrected formulations, and states the corresponding amendment needed in the Lean headline.

1 Source checked

The source examined here is the original submission, not a repository paraphrase:

> Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1, 16 December 2025.

The local archival copies are `paper2.pdf` and the submitted TeX source `.tmp/arxiv/2512.14858.tar`. As of 12 July 2026, arXiv lists no later revision. Theorem 1.2 assumes $a, b \geq 0$ and $\beta \geq 1$. It claims boundedness when $0 < m < 1$, and boundedness plus global existence when $m = 1$ and

$$\chi_0 < \frac{2(2\beta - 1)}{\max\{2, \gamma N\}}. \tag{1}$$

Remark 1.4(2) also says that a and b may vanish. No preceding convention excludes the mixed case $a > 0$, $b = 0$.

2 Counterexample

Let $a > 0$, $b = 0$, and let the initial datum be the positive constant $u_0(x) = c$. On any smooth bounded Neumann domain, set

$$u(t, x) = ce^{at}, \qquad v(t, x) = \frac{\nu}{\mu} u(t, x)^{\gamma}. \tag{2}$$

Both functions are spatially constant. Therefore all spatial derivatives and the chemotaxis divergence vanish. The elliptic equation for v holds at every time, while the parabolic equation reduces to

$$u_t = au. \tag{3}$$

Thus (2) is a positive global classical solution for every $m > 0$, every $\beta \geq 0$, and every $\chi_0 \in \mathbb{R}$. However,

$$\|u(t, \cdot)\|_{\infty} = ce^{at} \longrightarrow \infty. \tag{4}$$

Consequently, both parts of the printed theorem fail in the branch $a > 0$, $b = 0$. Part (2) already fails for $m = 1$, $\beta = 1$, and $\chi_0 = 0$, which satisfy (1).

There is also a solution-independent formulation of the contradiction. For every positive classical Neumann solution with $b = 0$, integration of the parabolic equation gives

$$M'(t) = aM(t), \qquad M(t) := \int_\Omega u(t,x)\,dx. \tag{5}$$

Positivity gives $M(t) > 0$, so $a > 0$ forces unbounded mass. On a bounded unit-volume domain,

$$M(t) \le \|u(t,\cdot)\|_\infty. \tag{6}$$

Hence an eventually bounded positive global solution cannot exist in this branch, independently of uniqueness or of a preferred explicit solution.

3 Where the proof loses the hypothesis

The missing parameter restriction is visible in both parts of Section 4.

1. Proposition 2.4 supplies a uniform mass bound only in the two cases

$$a = b = 0, \qquad \text{or} \qquad a > 0 \text{ and } b > 0. \tag{7}$$

2. In Section 4.2, the proof of Theorem 1.2(1) drops the nonpositive term $-b\int_\Omega u^{p+\alpha}$ and later invokes Lemma 4.1 together with Proposition 2.4 to replace a mass-dependent quantity by a time-independent constant. Proposition 2.4 supplies no such constant when $a > 0$, $b = 0$.

3. In Section 4.3, the proof of Theorem 1.2(2) reduces to $b = 0$ and obtains an inequality of the form

$$\frac{1}{p}\frac{d}{dt}\int_\Omega u^p \le -\int_\Omega u^p + C\left(\int_\Omega u\right)^p. \tag{8}$$

   It then claims a uniform-in-time L^p bound by Gronwall. That conclusion requires a uniform mass bound. In the omitted mixed branch, (5) makes the forcing term in (8) grow exponentially, so Gronwall yields growth rather than a uniform bound.

4. The appeal to Theorem 1.1 for $\chi_0 \le 0$ does not repair the mixed case. Theorem 1.1 itself treats only the two branches in (7).

4 Corrected parameter statement

The exact correction directly supported by the written proof is

$$(a = b = 0) \quad \text{or} \quad (a > 0 \text{ and } b > 0). \tag{9}$$

A slightly stronger and mathematically natural correction is

$$a = 0 \quad \text{or} \quad b > 0. \tag{10}$$

Under the standing assumptions $a, b \ge 0$, condition (10) is equivalent to excluding exactly

$$a > 0 \quad \text{and} \quad b = 0. \tag{11}$$

The additional case $a = 0$, $b > 0$ needs only the elementary mass estimate

$$M'(t) = -b \int_\Omega u^{1+\alpha} \le 0, \tag{12}$$

so it provides the same uniform mass input used in Section 4.

Recommended amendment. Assume $\beta \ge 1$ and either $a = 0$ or $b > 0$. If $0 < m < 1$, then (1.23) holds. If $m = 1$ and (1) holds, then (1.23) holds and $T_{\max}(u_0) = \infty$.

If the authors do not wish to add the elementary $a = 0$, $b > 0$ mass case, the first sentence should instead assume (9). Remark 1.4(2) should then say that both parameters may vanish simultaneously, rather than suggesting all mixed nonnegative cases.

5 Lean amendment

The repository contains both a concrete and a parameter-general interval-domain refutation in `ShenWork/Paper2/IntervalDomainTheorem12Refutation.lean`:

```
not_Theorem_1_2_intervalDomain_when_a_pos_b_zero
not_Theorem_1_2_intervalDomain_of_a_pos_b_zero
```

The proofs contain no `sorry` and depend only on `propext`, `Classical.choice`, and `Quot.sound`.

The existing Lean definition has a second, independent fidelity problem in part (1): it asks only for some bounded local solution on some positive finite horizon. The paper fixes the unique maximal solution and asserts boundedness on its full maximal interval. A faithful headline should use the existing continuation objects in `ShenWork/PDE/IntervalDomainExistence.lean`:

```
ReachableClassicalHorizon
ReachableArbitrarilyLong
FiniteContinuationAlternativeBranch
StandardContinuationAlternative
```

A bare existential local witness does not formalize (1.23). The amended headline must distinguish a finite maximal-time alternative, available only when reachable horizons are bounded above, from a genuine glued global branch when they are unbounded.

Theorem 1.3 is unaffected by this particular $a > 0$, $b = 0$ counterexample, because its original statement explicitly assumes $a, b > 0$. A separate exponent-domain defect in the proof of Theorem 1.3(iv) is documented in `paper2_theorem13_case_iv_amendment.pdf`.

Reference

L. Chen, I. Ruau, and W. Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1 (2025), `https://arxiv.org/abs/2512.14858`.